

```

timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module instr_mem(
    input wire clk,
    input wire rst_n, // active low reset
    input wire [31:0] i_addr, // instruction address
    input wire i_stb, // request for instruction
    output reg o_ack, // acknowledge signal
    output reg [31:0] o_data // instruction code
);

reg [31:0] memory [1023:0];

initial
begin
    $readmemh("../rtl/riscvtest.txt", memory, 0, 20); // read instruction from the
.mem file (byte wise)
end

always @(*)
begin
    if(~rst_n)
    begin
        o_ack = 1'b0;
        o_data = 32'd0;
    end
    else if(i_stb && ( (i_addr & 2'b11) == 2'b00 )) // checking if the stub signal
is high and address is 4-byte aligned
    begin
        o_ack <= 1'b1; // acknowledge that the instruction is on the bus.
        o_data <= memory [i_addr>>2]; // instructions are present byte wise and are
in little-endian format
    end
    else
    begin
        if( (i_addr & 2'b11) != 2'b00 ) begin
            $display("\nINSTRUCTION MEMORY: Address %h is not 4-byte aligned!",
i_addr);
        end
        o_ack <= 1'b0;
        o_data <= 32'd0;
    end
end
end

```

endmodule